

Exercise Sheet 2

Advanced Microeconomics: Economic Decision Making

Question 1

Let $z_L(w, q)$ be conditional labour demand, differentiable, h.d.0 in $w = (w_1, \dots, w_n)$, with elasticities $\varepsilon_{Lj} = \partial \ln z_L / \partial \ln w_j$ and $\varepsilon_{LL} \leq 0$.

(a) Prove $\sum_j \varepsilon_{Lj} = 0$. [7]

(b) All input prices change by proportions $(g_1, \dots, g_n)$. Prove, to first order,

$$d \ln z_L = \sum_{j \neq L} \varepsilon_{Lj} (g_j - g_L),$$

so that only *relative* price changes move the input mix. [10]

(c) An economy indexes wages to a consumer price index containing import prices that never enter any domestic firm's revenues. Using (b), state the exact condition under which such indexation leaves every firm's input mix unchanged, and the sign of the labour-mix response when the condition fails. [8]

Question 2

A CRS competitive industry has labour cost share $s_L = 0.20$, other-input share $s_O = 0.80$, substitution elasticity $\sigma = 0.5$, and faces industry demand of elasticity $\varepsilon_D = -0.5$. In a shock year, other input costs rise $g_O = 8\%$; indexation raises wages $g_w = 12\%$.

(a) Compute, to first order, the output price, output, and total labour responses for the cost shock alone ($g_w = 0$). [9]

(b) Repeat with the indexation added, and compute the policy's marginal contribution to each. [10]

(c) Compute the labour-share-of-revenue response in both runs and give a one-sentence reading of the policy's marginal effect on it. [6]

Question 3

Technology $q = z_1^\alpha z_2^{1-\alpha}$, $\alpha \in (0, 1)$.

(a) Solve the CMP: derive $z(w, q)$ and show $c(w, q) = q \kappa w_1^\alpha w_2^{1-\alpha}$ with $\kappa = \alpha^{-\alpha} (1 - \alpha)^{-(1-\alpha)}$. [12]

- (b) Verify Shephard's lemma, and show the cost shares equal the exponents. [7]
- (c) Input prices change by proportions $(g_1, g_2) = (12\%, 3\%)$ with $\alpha = 0.25$. Compute $d \ln c$ to first order and interpret each term. [6]